Namal University, Mianwali

Design Report

Project: Point-of-Sale System

Course: CSC-225 – Software Engineering

Department: Computer Science

Semester: Fall 2025

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Submission Date: 18 Jan 2026

Contents

1 Introduction

1.1 Purpose

This document explains the design of our Point of Sale (POS) system. It shows how the functional requirements are translated into system components, interfaces, workflows, and database structures. This helps guide the system development and testing.

1.2 Project Overview

Our POS system is designed for restaurants or cafes to manage orders, billing, kitchen operations, and reporting. It replaces manual order tracking with a digital system that supports order placement, payment, kitchen preparation, and notifications to staff. The system also manages discounts, complementary items, and generates sales reports.

1.3 Design Approach

The design of the POS system follows these ideas:

- Independent modules for easier maintenance.
- Every component links to functional requirements.
- Interfaces are simple for fast and clear operation.
- Real-time communication between order-taking, kitchen, and cashier.

2 Design Assumptions and Constraints

2.1 Design Assumptions

- Users have access to a computer or tablet with the POS software.
- Tables, menu items, and staff information are preloaded.
- Internet or local network is available for notifications and data sync.
- User roles (Admin, Waiter, Cashier, Kitchen Staff) are predefined.
- Orders, payments, and reports are handled in the system only.

2.2 Design Constraints

- The system must follow restaurant policies and local data rules.
- Software runs on local devices or restaurant servers.
- Must handle multiple orders and users simultaneously.
- Payment and sensitive data must be securely stored.

3 Key Design Decisions

3.1 Architectural Decisions

The system is designed with a modular architecture to separate responsibilities:

- **Presentation Layer:** Interfaces for Waiters, Kitchen Staff, Cashiers, and Admins. Handles order entry, notifications, and dashboards.
- **Application Logic Layer:** Contains main services: OrderManagement, BillingService, KitchenService, NotificationService, UserManagementService. Handles business logic like order processing, discounts, complementary items, and payment processing.
- **Data Access Layer:** Communicates with databases: Orders, Menu Items, Tables, Complementary Items, Order Status. Ensures correct storage and retrieval of data.

This separation helps with maintenance, testing, and scaling.

3.2 Security Design Decisions

- Role-based access control to ensure only allowed user to login.
- Authentication with secure passwords.
- Session management with timeouts.
- Audit logs for important actions (order creation, payment, user management).

3.3 User Interface Design Decisions

- **Waiter Dashboard:** Table selection, order placement, modification, and serving confirmation.
- **Kitchen Dashboard:** Shows pending orders, allows marking orders as "In Preparation" or "Prepared."
- **Cashier Dashboard:** Billing, payment processing, and receipt generation.
- **Admin Dashboard:** Reports, menu management and Account management..

4 Design Diagrams

This section presents all the design diagrams of the POS system, including Data Flow Diagrams (DFDs), Component Diagram, Class Diagrams, Sequence Diagrams, and Activity Diagrams. Each diagram illustrates the system structure, interactions, and operational workflows.

4.1 Data Flow Diagrams (DFD)

4.2 Class Diagrams

To improve clarity and modularity, the class diagram is divided into two logical parts based on system responsibilities.

4.2.1 Class Diagram 1: Core Operations

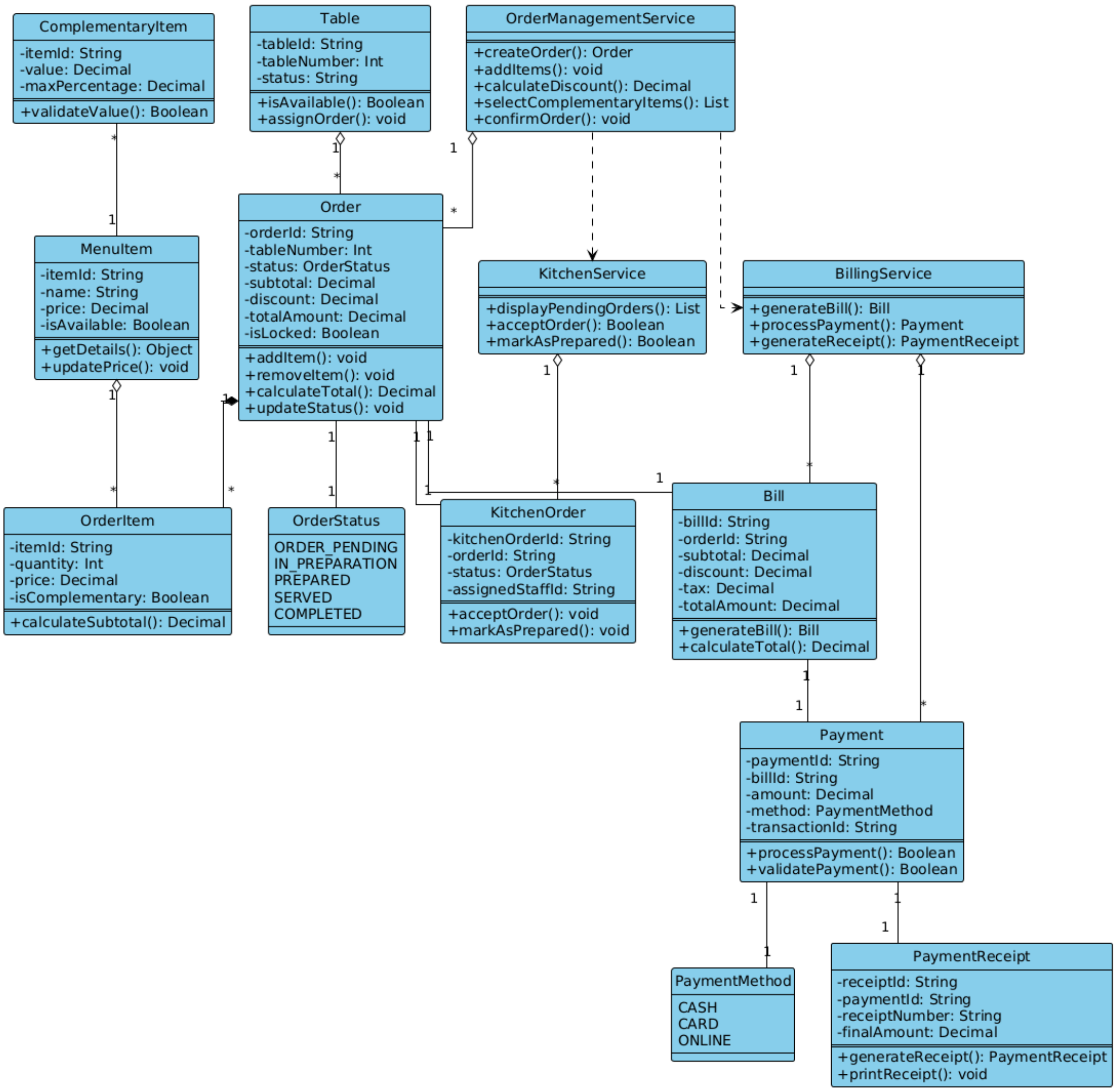


Figure 1: Class Diagram – Core Operations

This diagram contains tightly coupled operational classes responsible for the core functionality of the POS system, including:

- **Order Management**: Table, Order, OrderItem, ComplementaryItem
- **Kitchen Operations**: Kitchen, KitchenOrder
- **Billing and Payment**: Bill, Payment, PaymentReceipt, PaymentGateway
- Related service classes, enumerations, and status management

4.2.2 Class Diagram 2: Administration & Support

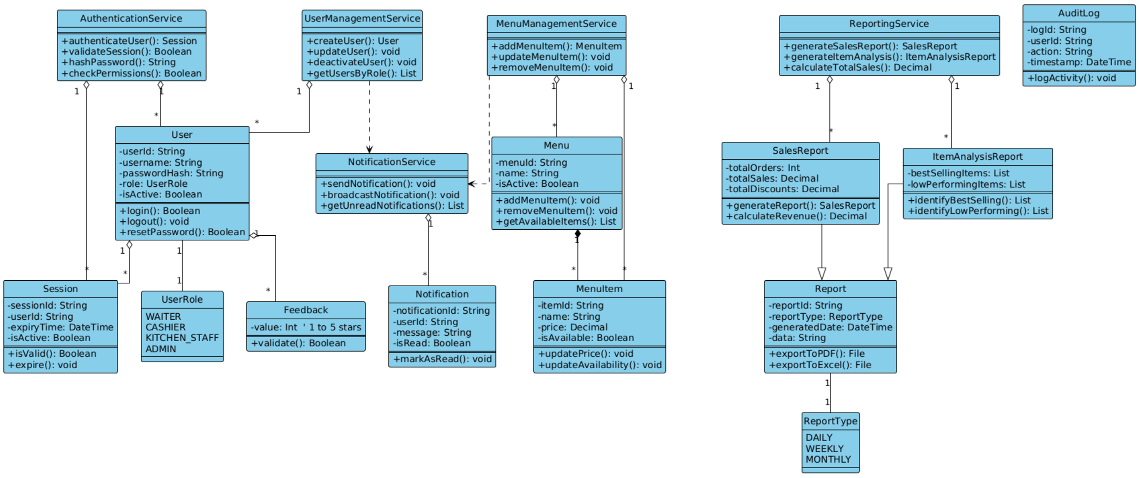


Figure 2: Class Diagram – Administration and Support

This diagram focuses on administrative and support functionalities of the system, including:

- User Management and Authentication
- Menu Management
- Feedback System
- Reporting and Analytics (Sales Reports, Item Performance)
- Audit Logging
- Notification Services

4.3 Sequence Diagrams

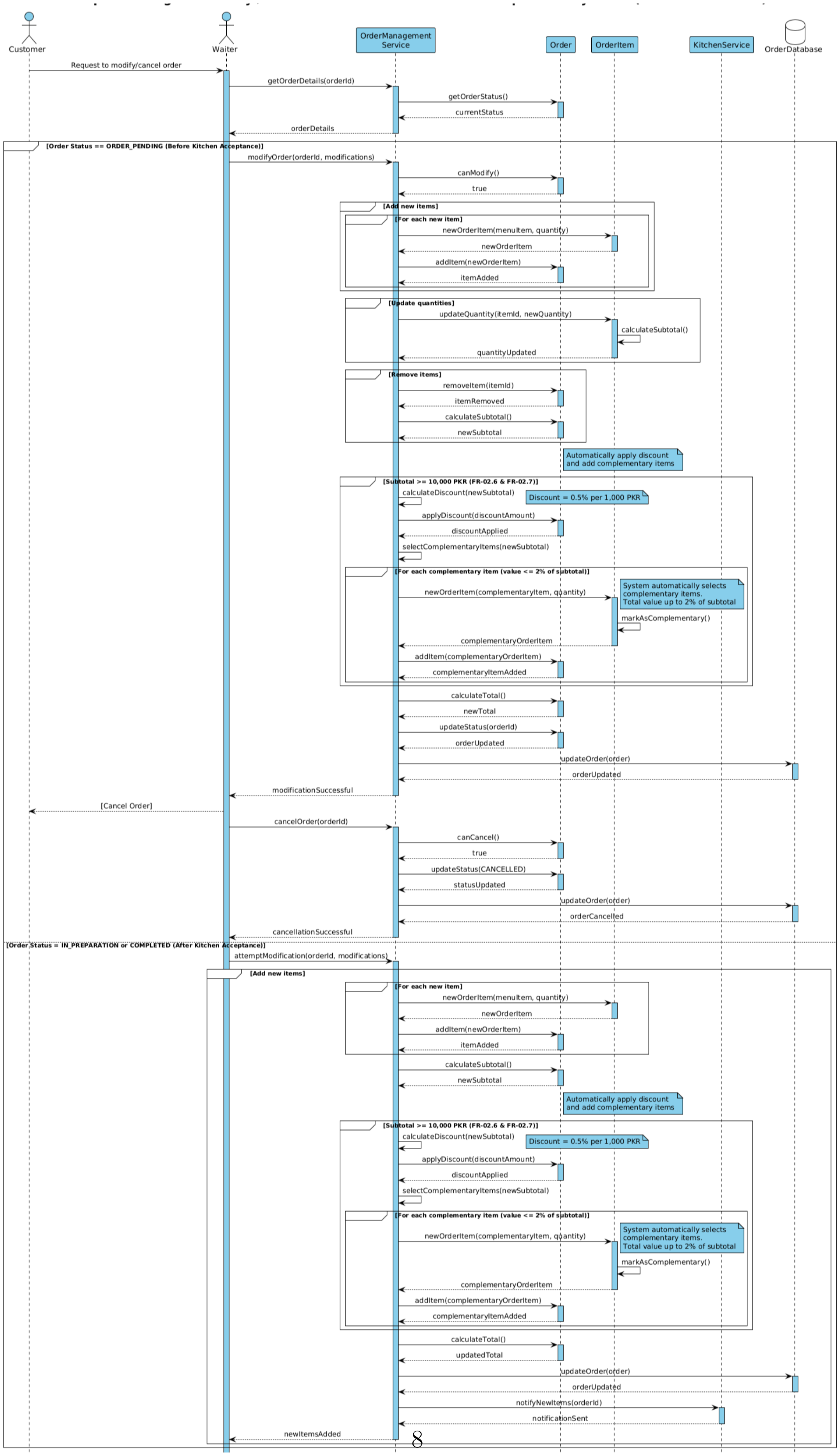


Figure 3: Sequence Diagram – Modify / Cancel Order

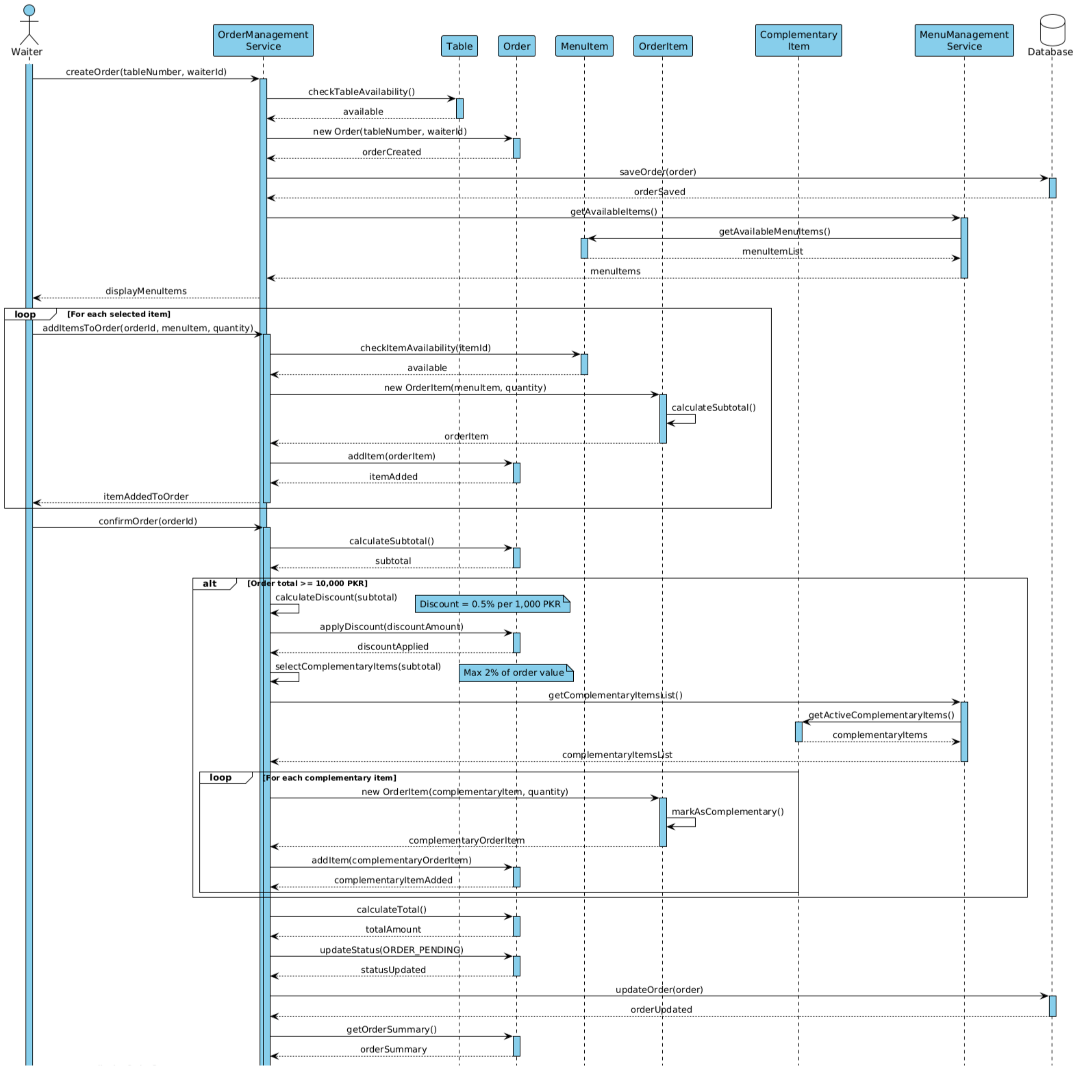


Figure 4: Sequence Diagram – Place Order

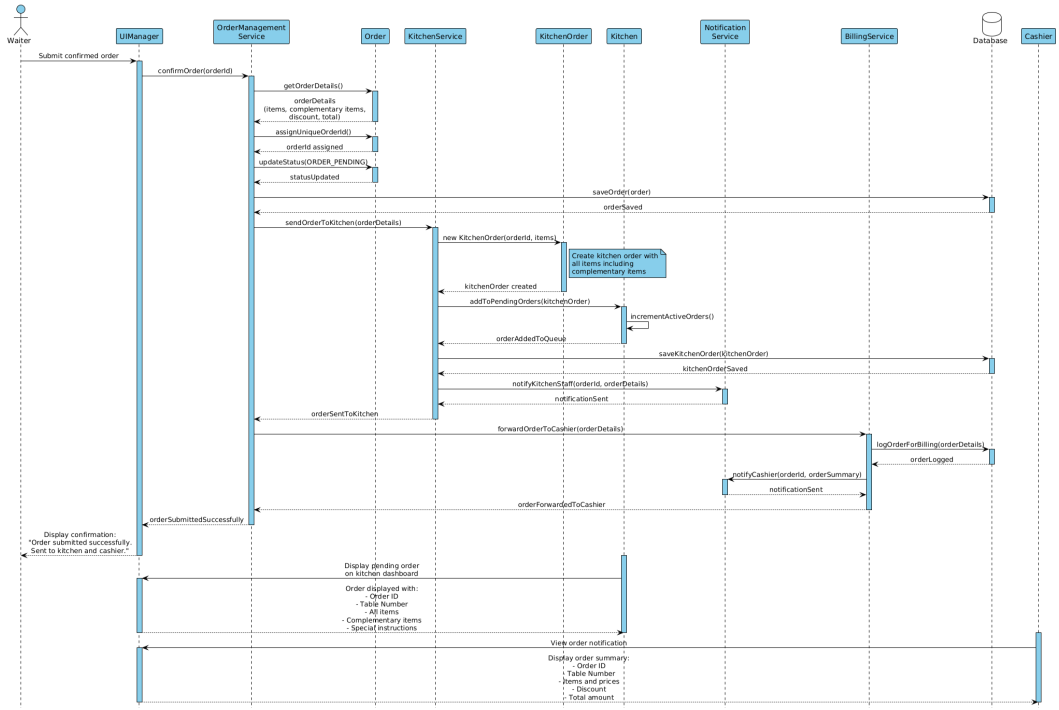
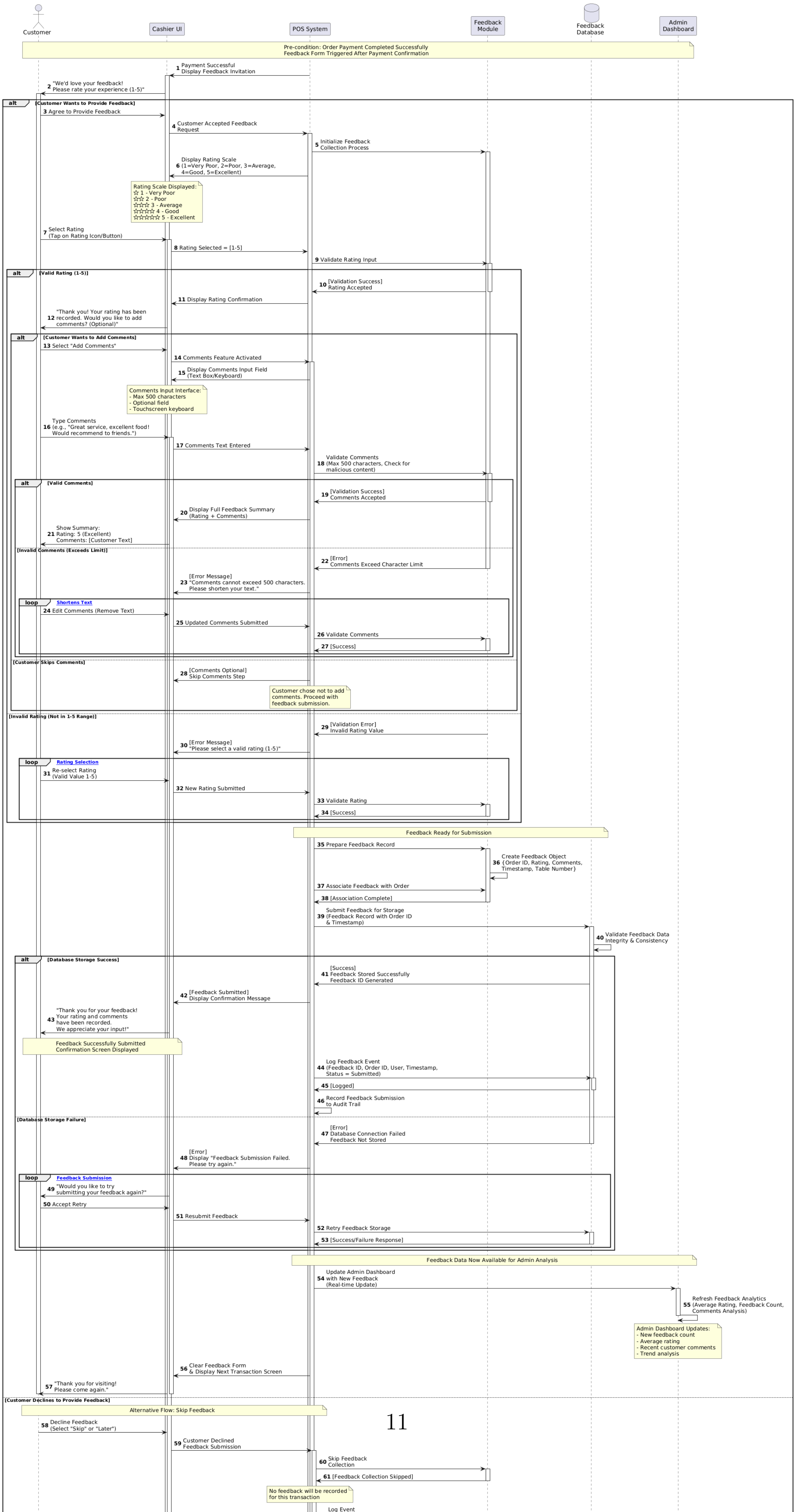
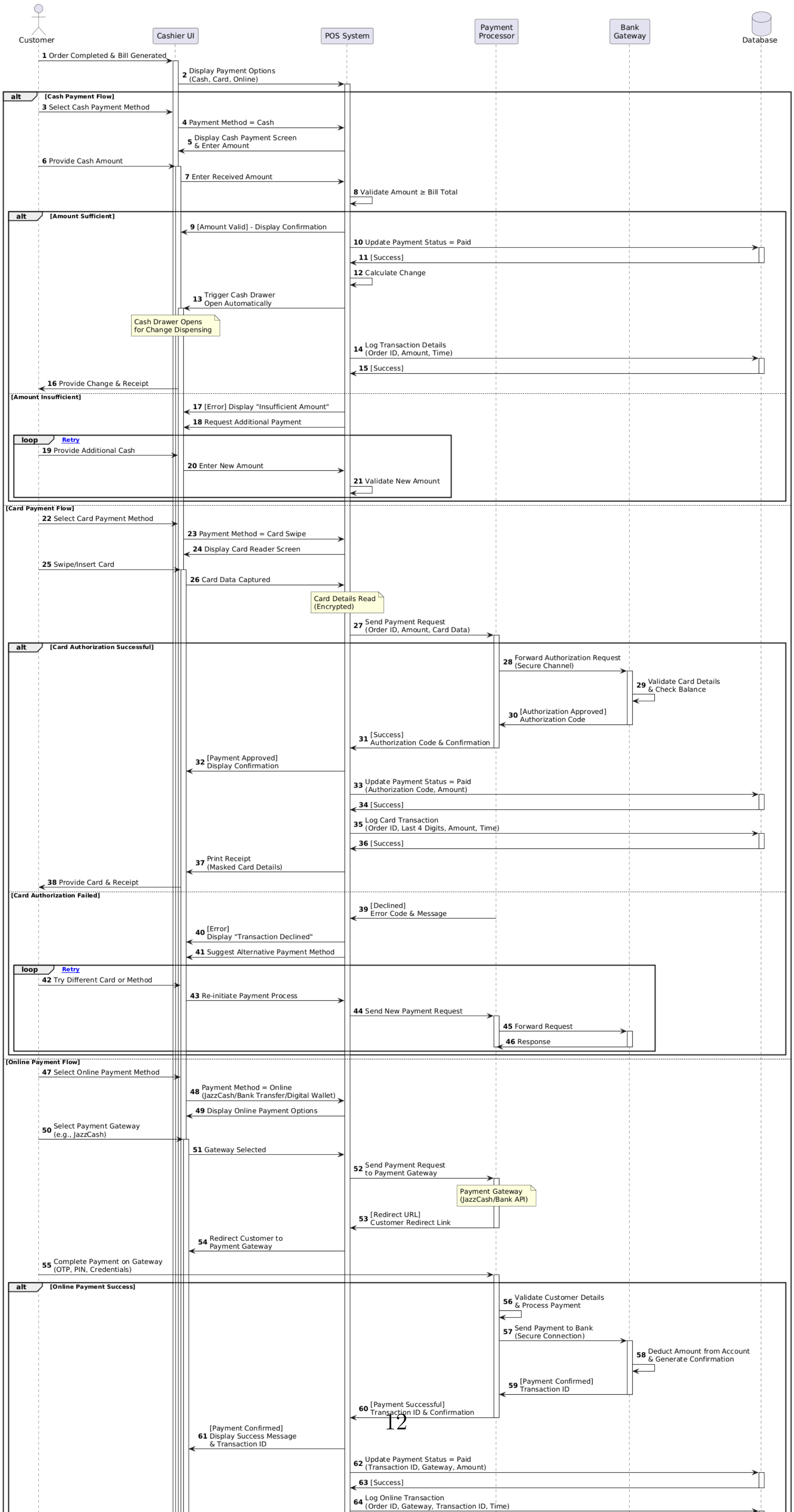
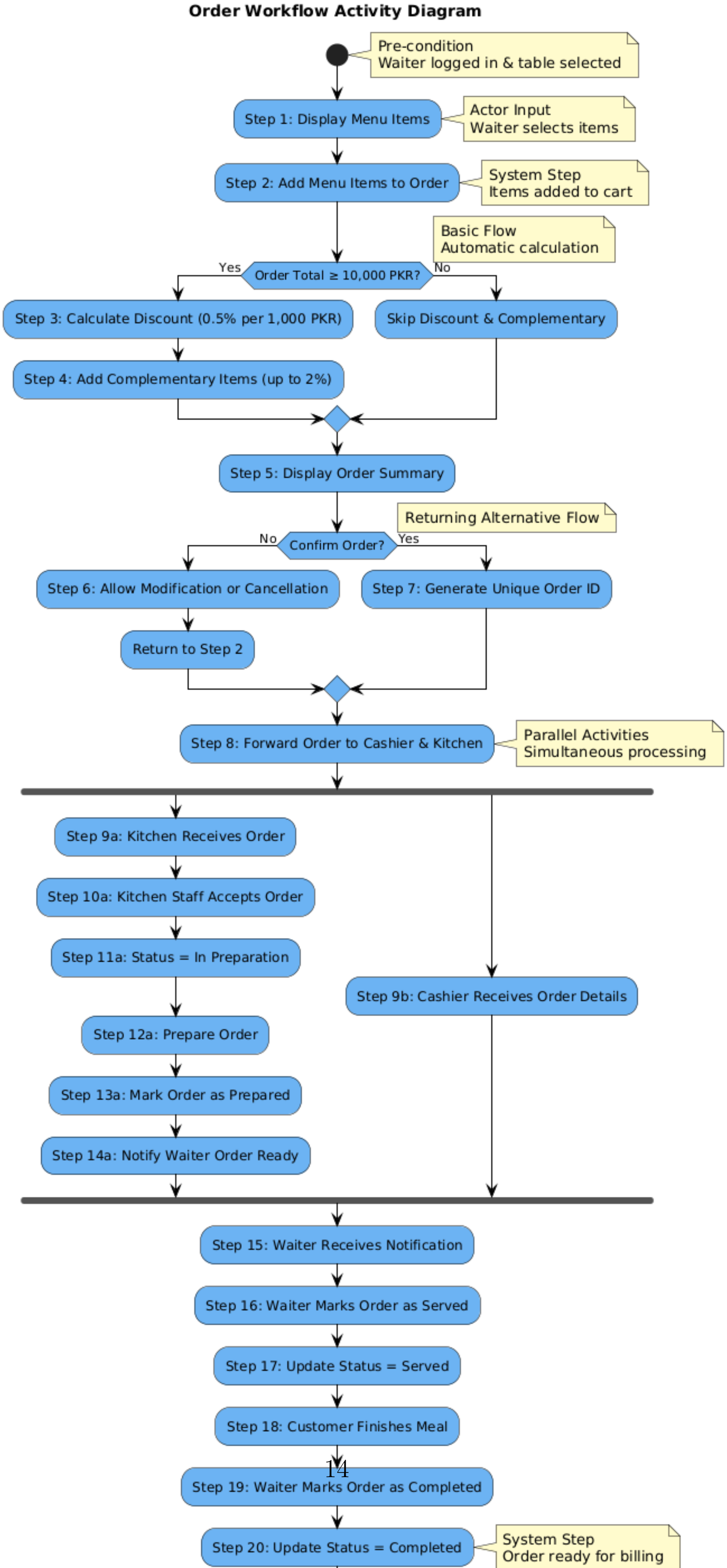


Figure 5: Sequence Diagram – Send Order to Kitchen


Customer
Cashier UI
POS System
Feedback Module
Feedback Database
Admin Dashboard
Pre-condition: Order Payment Completed Successfully
Feedback Form Triggered After Payment Confirmation
1 Payment Successful
Display Feedback Invitation
2 "We'd love your feedback!
Please rate your experience (1-5)"
alt [Customer Wants to Provide Feedback]
3 Agree to Provide Feedback
4 Customer Accepted Feedback
Request
5 Initialize Feedback
Collection Process
6 Display Rating Scale
(1=Very Poor, 2=Poor, 3=Average,
4=Good, 5=Excellent)
Rating Scale Displayed:
☆ 1 - Very Poor
☆☆ 2 - Poor
☆☆☆ 3 - Average
☆☆☆☆ 4 - Good
☆☆☆☆☆ 5 - Excellent
7 Select Rating
(Tap on Rating Icon/Button)
8 Rating Selected = [1-5]
9 Validate Rating Input
alt [Valid Rating (1-5)]
10 [Validation Success]
Rating Accepted
11 Display Rating Confirmation
12 "Thank you! Your rating has been
recorded. Would you like to add
comments? (Optional)"
alt [Customer Wants to Add Comments]
13 Select "Add Comments"
14 Comments Feature Activated
15 Display Comments Input Field
(Text Box/Keyboard)
Comments Input Interface:
- Max 500 characters
- Optional field
- Touchscreen keyboard
16 Type Comments
(e.g., "Great service, excellent food!
Would recommend to friends.")
17 Comments Text Entered
18 Validate Comments
(Max 500 characters, Check for
malicious content)
alt [Valid Comments]
19 [Validation Success]
Comments Accepted
20 Display Full Feedback Summary
(Rating + Comments)
21 Show Summary:
Rating: 5 (Excellent)
Comments: [Customer Text]
[Invalid Comments (Exceeds Limit)]
22 [Error]
Comments Exceed Character Limit
23 [Error Message]
"Comments cannot exceed 500 characters.
Please shorten your text."
loop Shortens Text
24 Edit Comments (Remove Text)
25 Updated Comments Submitted
26 Validate Comments
27 [Success]
[Customer Skips Comments]
28 [Comments Optional]
Skip Comments Step
Customer chose not to add
comments. Proceed with
feedback submission.
[Invalid Rating (Not in 1-5 Range)]
29 [Validation Error]
Invalid Rating Value
30 [Error Message]
"Please select a valid rating (1-5)"
loop Rating Selection
31 Re-select Rating
(Valid Value 1-5)
32 New Rating Submitted
33 Validate Rating
34 [Success]
Feedback Ready for Submission
35 Prepare Feedback Record
36 Create Feedback Object
{Order ID, Rating, Comments,
Timestamp, Table Number}
37 Associate Feedback with Order
38 [Association Complete]
39 Submit Feedback for Storage
(Feedback Record with Order ID
& Timestamp)
40 Validate Feedback Data
Integrity & Consistency
alt [Database Storage Success]
41 [Success]
Feedback Stored Successfully
Feedback ID Generated
42 [Feedback Submitted]
Display Confirmation Message
43 "Thank you for your feedback!
Your rating and comments
have been recorded.
We appreciate your input!"
Feedback Successfully Submitted
Confirmation Screen Displayed
44 Log Feedback Event
(Feedback ID, Order ID, User, Timestamp,
Status = Submitted)
45 [Logged]
46 Record Feedback Submission
to Audit Trail
[Database Storage Failure]
47 [Error]
Database Connection Failed
Feedback Not Stored
48 [Error]
Display "Feedback Submission Failed.
Please try again."
loop Feedback Submission
49 "Would you like to try
submitting your feedback again?"
50 Accept Retry
51 Resubmit Feedback
52 Retry Feedback Storage
53 [Success/Failure Response]
Feedback Data Now Available for Admin Analysis
54 Update Admin Dashboard
with New Feedback
(Real-time Update)
55 Refresh Feedback Analytics
(Average Rating, Feedback Count,
Comments Analysis)
Admin Dashboard Updates:
- New feedback count
- Average rating
- Recent customer comments
- Trend analysis
56 Clear Feedback Form
& Display Next Transaction Screen
57 "Thank you for visiting!
Please come again."
[Customer Declines to Provide Feedback]
Alternative Flow: Skip Feedback
58 Decline Feedback
(Select "Skip" or "Later")
59 Customer Declined
Feedback Submission
60 Skip Feedback
Collection
61 [Feedback Collection Skipped]
No feedback will be recorded
for this transaction
Log Event


Customer
Cashier UI
POS System
Payment Processor
Bank Gateway
Database
1 Order Completed & Bill Generated
2 Display Payment Options (Cash, Card, Online)
alt [Cash Payment Flow]
3 Select Cash Payment Method
4 Payment Method = Cash
5 Display Cash Payment Screen & Enter Amount
6 Provide Cash Amount
7 Enter Received Amount
8 Validate Amount ≥ Bill Total
alt [Amount Sufficient]
9 [Amount Valid] - Display Confirmation
10 Update Payment Status = Paid
11 [Success]
12 Calculate Change
13 Trigger Cash Drawer Open Automatically
Cash Drawer Opens for Change Dispensing
14 Log Transaction Details (Order ID, Amount, Time)
15 [Success]
16 Provide Change & Receipt
[Amount Insufficient]
17 [Error] Display "Insufficient Amount"
18 Request Additional Payment
loop Retry
19 Provide Additional Cash
20 Enter New Amount
21 Validate New Amount
[Card Payment Flow]
22 Select Card Payment Method
23 Payment Method = Card Swipe
24 Display Card Reader Screen
25 Swipe/Insert Card
26 Card Data Captured
Card Details Read (Encrypted)
27 Send Payment Request (Order ID, Amount, Card Data)
alt [Card Authorization Successful]
28 Forward Authorization Request (Secure Channel)
29 Validate Card Details & Check Balance
30 [Authorization Approved] Authorization Code
31 [Success] Authorization Code & Confirmation
32 [Payment Approved] Display Confirmation
33 Update Payment Status = Paid (Authorization Code, Amount)
34 [Success]
35 Log Card Transaction (Order ID, Last 4 Digits, Amount, Time)
36 [Success]
37 Print Receipt (Masked Card Details)
38 Provide Card & Receipt
[Card Authorization Failed]
39 [Declined] Error Code & Message
40 [Error] Display "Transaction Declined"
41 Suggest Alternative Payment Method
loop Retry
42 Try Different Card or Method
43 Re-initiate Payment Process
44 Send New Payment Request
45 Forward Request
46 Response
[Online Payment Flow]
47 Select Online Payment Method
48 Payment Method = Online (JazzCash/Bank Transfer/Digital Wallet)
49 Display Online Payment Options
50 Select Payment Gateway (e.g., JazzCash)
51 Gateway Selected
52 Send Payment Request to Payment Gateway
Payment Gateway (JazzCash/Bank API)
53 [Redirect URL] Customer Redirect Link
54 Redirect Customer to Payment Gateway
55 Complete Payment on Gateway (OTP, PIN, Credentials)
alt [Online Payment Success]
56 Validate Customer Details & Process Payment
57 Send Payment to Bank (Secure Connection)
58 Deduct Amount from Account & Generate Confirmation
59 [Payment Confirmed] Transaction ID
60 [Payment Successful] Transaction ID & Confirmation
61 [Payment Confirmed] Display Success Message & Transaction ID
62 Update Payment Status = Paid (Transaction ID, Gateway, Amount)
63 [Success]
64 Log Online Transaction (Order ID, Gateway, Transaction ID, Time)

4.4 Activity Diagrams



1. Order Workflow

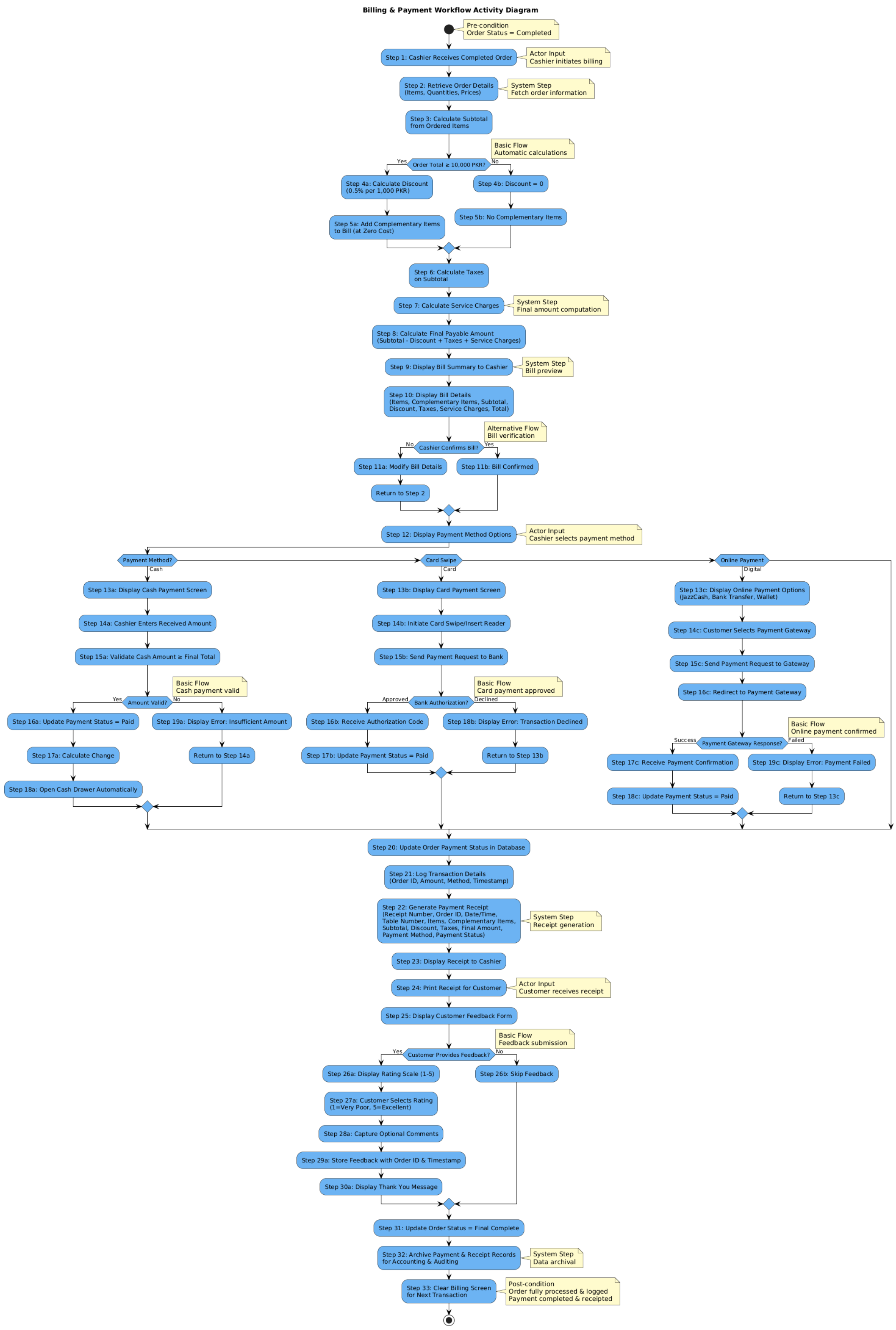


Figure 9: Activity Diagram – Billing and Payment Workflow

2. Billing and Payment Workflow

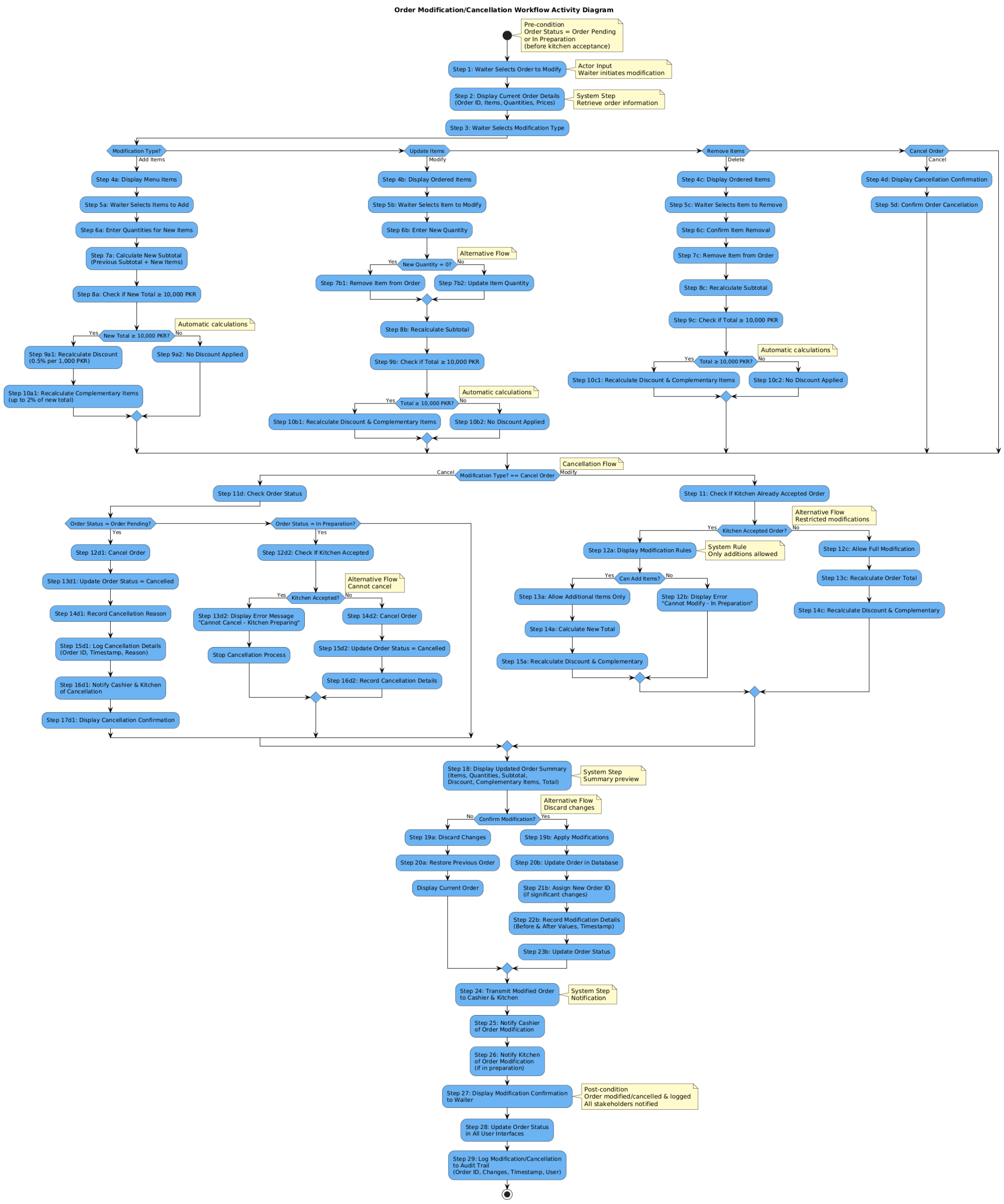


Figure 10: Activity Diagram – Order Modification / Cancellation Workflow

3. Order Modification / Cancellation Workflow

4.4.1 DFD Level 1

DFD Level 1 decomposes the POS system into four major functional processes. Each process is represented by a separate Level 1 diagram for clarity.

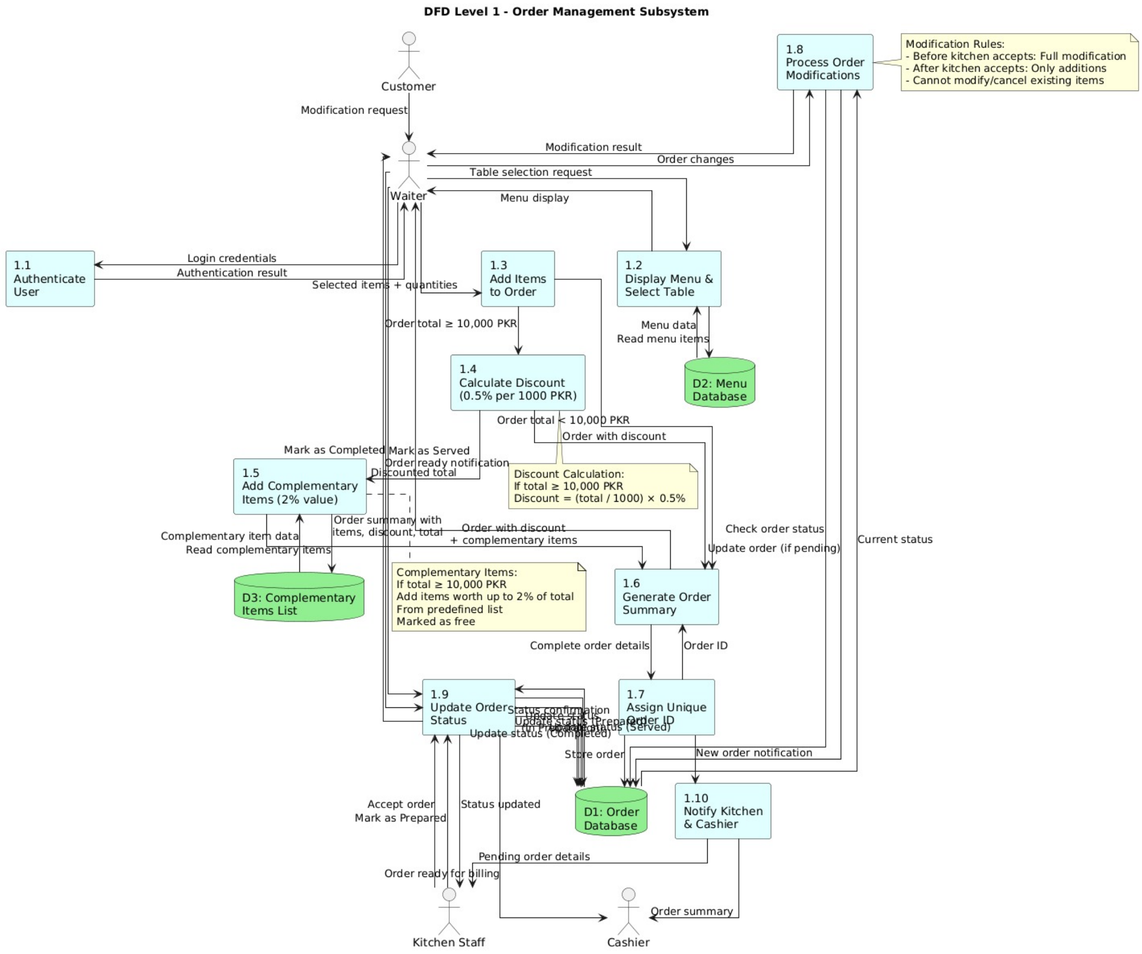


Figure 11: DFD Level 1 – Order Management

Order Management This diagram shows order-related activities including order creation, modification, confirmation, and status tracking.

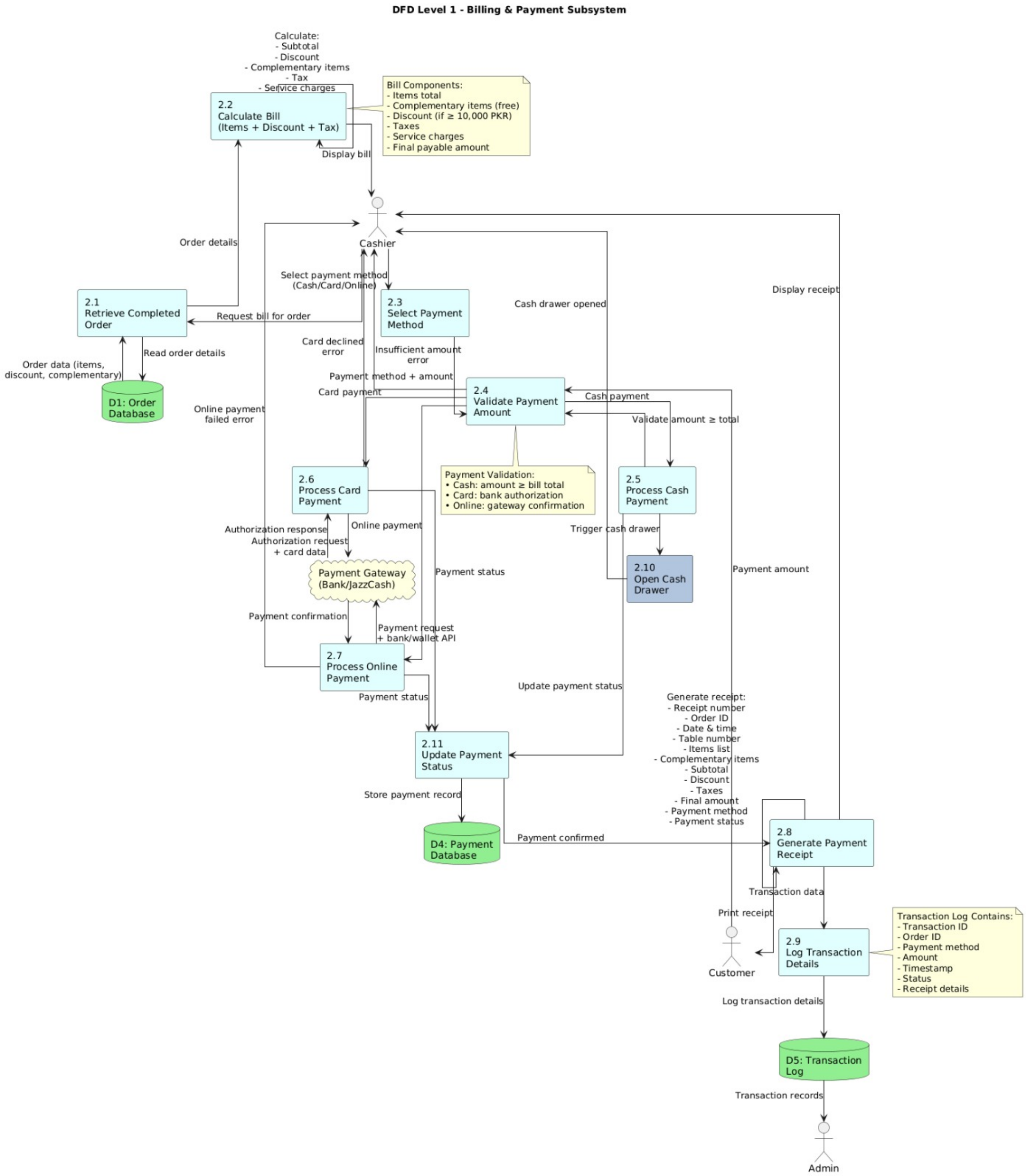


Figure 12: DFD Level 1 – Billing and Payment

Billing & Payment This diagram illustrates bill generation, payment processing, receipt generation, and transaction logging.

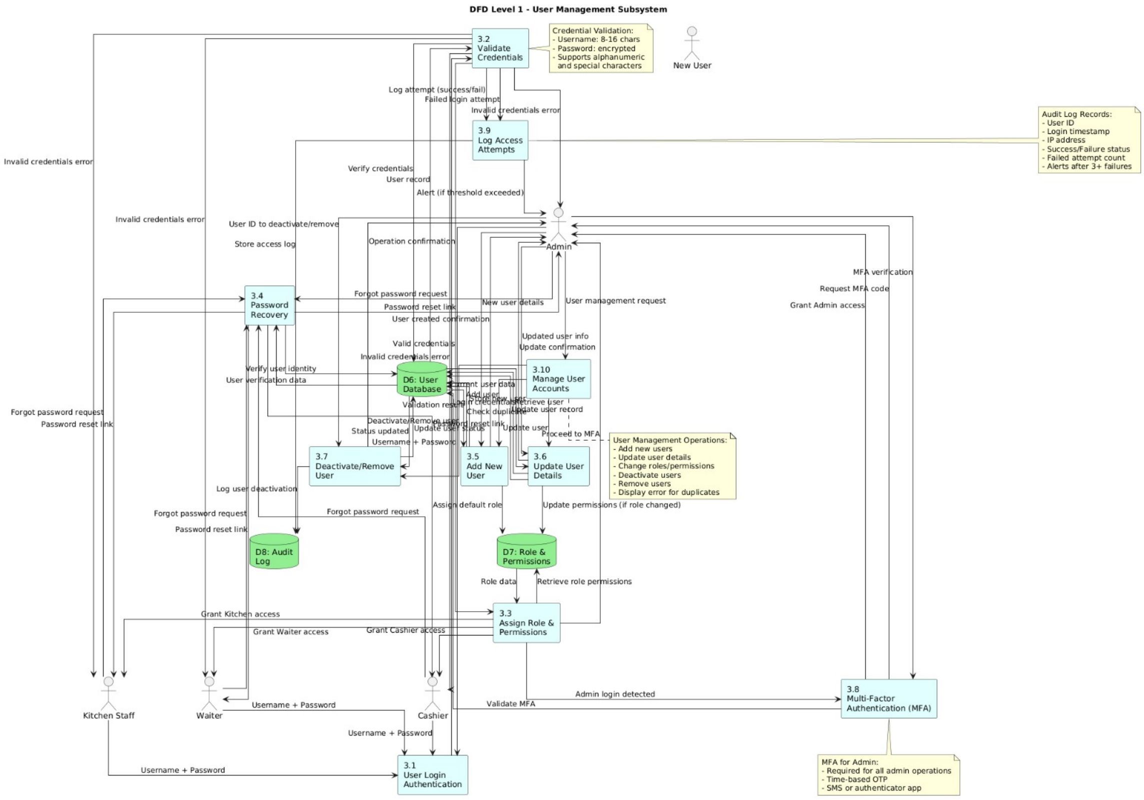
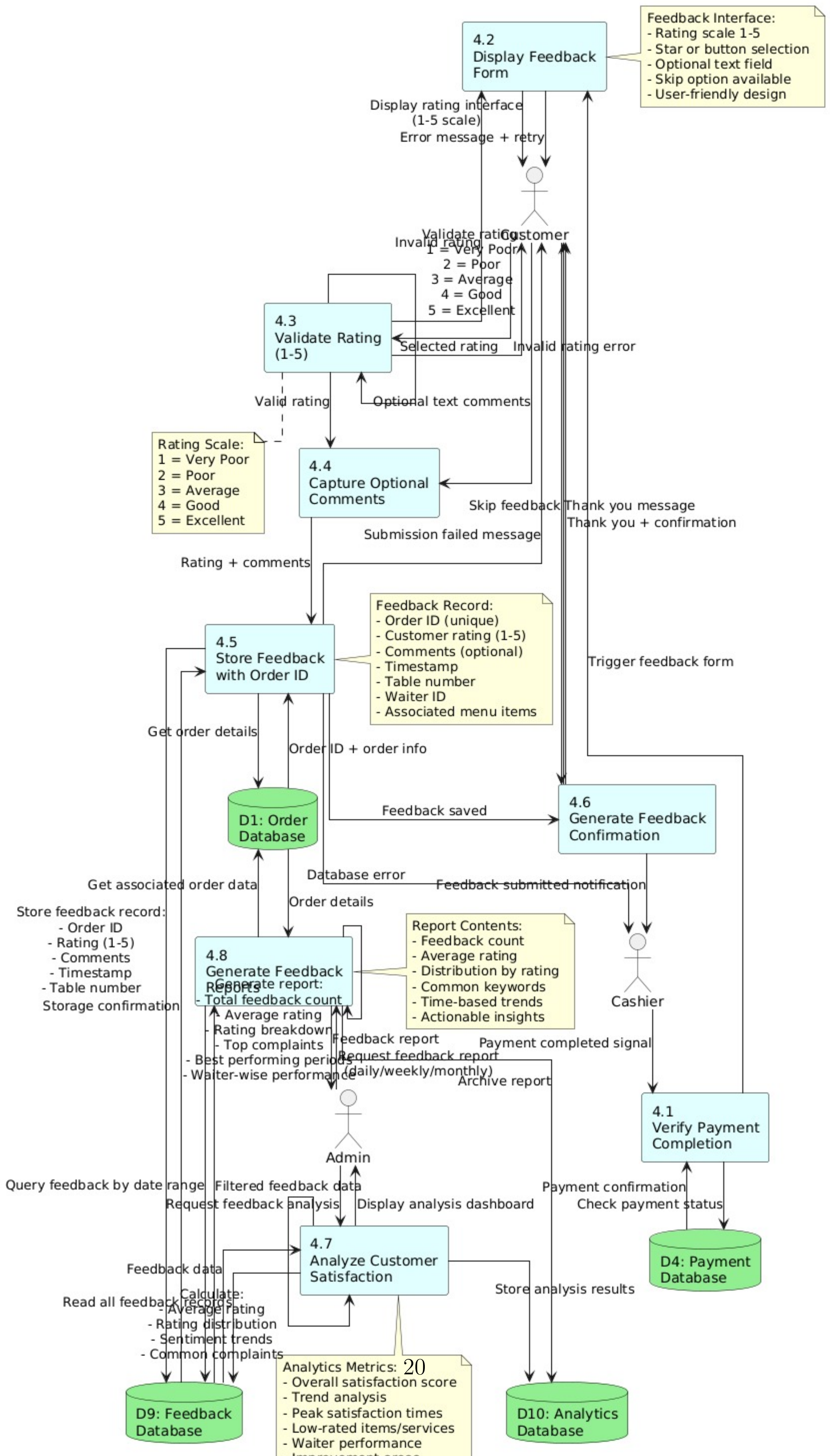


Figure 13: DFD Level 1 – User Management

User Management This diagram represents user authentication, role-based access control, and account management.

DFD Level 1 - Feedback Management Subsystem



Feedback Management

4.4.2 DFD Level 0 (Context Diagram)

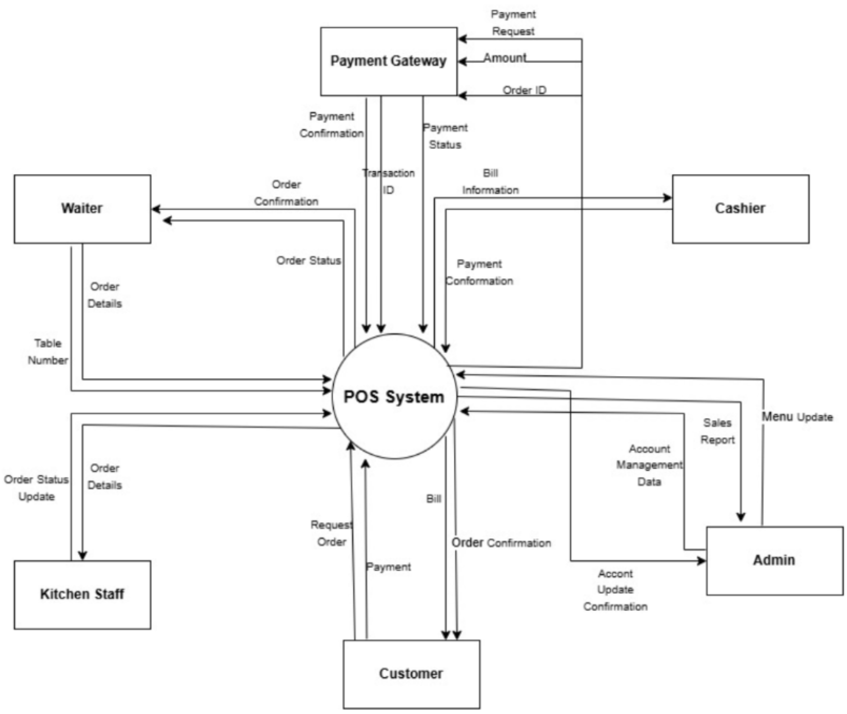


Figure 15: DFD Level 0 – Context Diagram

This diagram represents the POS system as a single process and shows its interaction with external entities such as Customers, Waiters, Kitchen Staff, Cashiers, and Admin.

This diagram shows feedback submission, rating capture, and feedback storage linked with completed orders.

4.4.3 DFD Level 2

DFD Level 2 is divided into two detailed subprocesses.

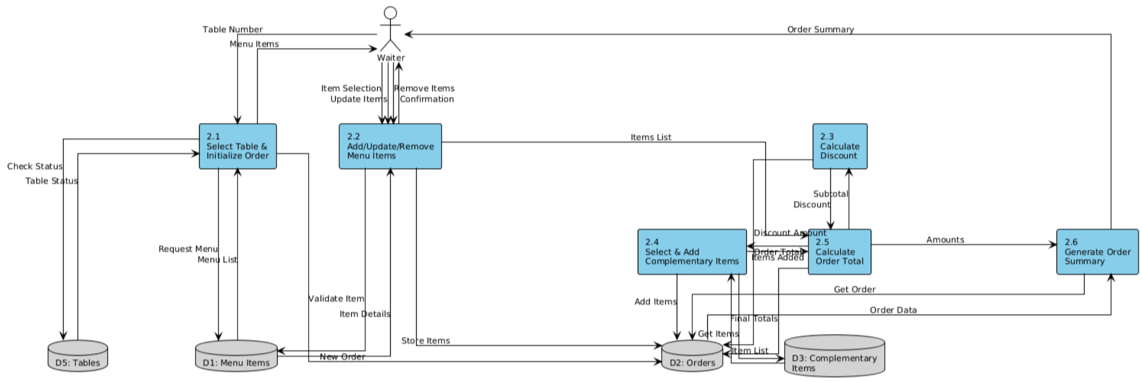


Figure 16: DFD Level 2.1 – Order Placement

2.1 Order Placement This diagram illustrates the detailed flow of placing an order, including table selection, menu item selection, discount and complementary item calculation, and order summary generation.

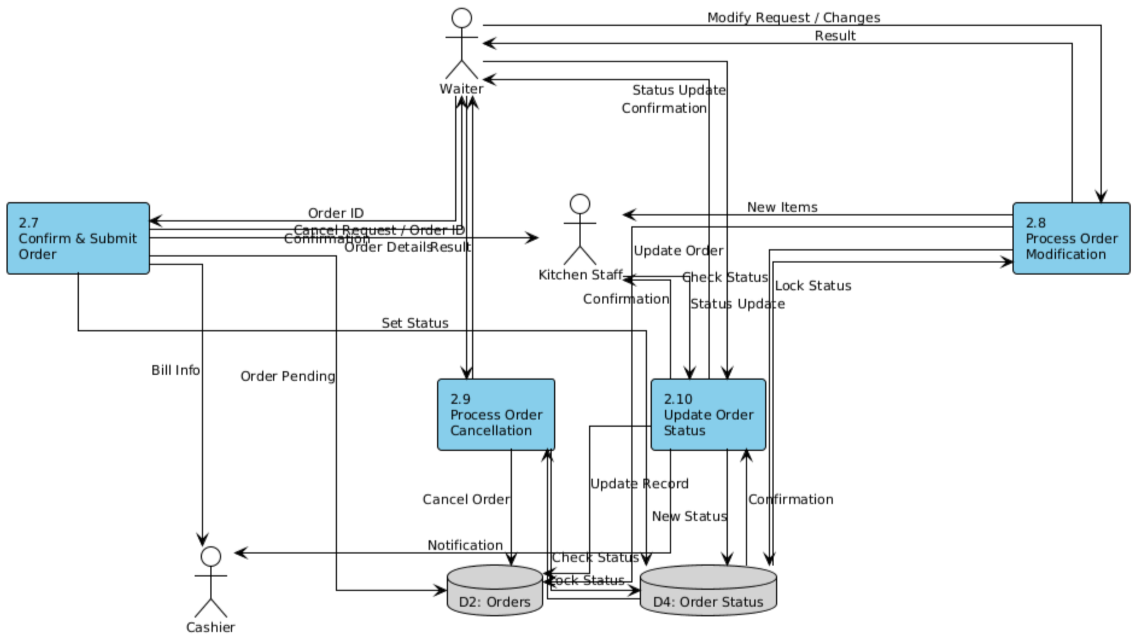


Figure 17: DFD Level 2.2 – Order Processing

2.2 Order Processing This diagram details order confirmation, transmission to kitchen and cashier, status management, order modification and cancellation rules, and lifecycle tracking.

4.5 Component Diagram

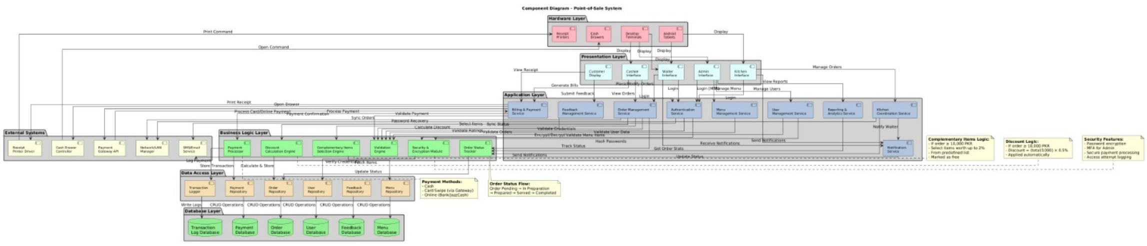


Figure 18: Component Diagram of POS System

The component diagram illustrates the high-level modular architecture of the POS system, showing interactions between components such as User Authentication, Order Management, Kitchen Operations, Billing and Payment, Reporting, and Administration.

Requirements-Design Traceability Table

FR ID	Functional Requirement	Design Mapping
FR-01.1	Allow authorized users to log in using username and password	Classes: UserManagementService; Components: User Authentication
FR-01.2	Validate entered credentials against stored user records	Classes: UserManagementService; Components: User Authentication
FR-01.3	Grant access based on authenticated user's role	Classes: UserManagementService, Admin, Waiter; Components: User Authentication
FR-01.4	Deny access and display error message if authentication fails	Classes: UserManagementService; Components: User Authentication
FR-01.5	Provide "Forgot Password" feature for password recovery	Classes: UserManagementService; Components: User Authentication

FR ID	Functional Requirement	Design Mapping
FR-02.1	Allow waiter to select table number for new order	DFD Level 2.1: Process 2.1 "Select Table & Initialize Order"; Sequence Diagram: Place Order; Activity Diagram: Order Workflow; Classes: OrderManagementService, Table, Waiter; Database: D5 Tables; Components: Order Placement
FR-02.2	Display available menu items after table selection	DFD Level 2.1: Process 2.1 accessing D1 Menu Items; Sequence Diagram: Place Order – displayMenuItems(); Classes: MenuManagementService, MenuItem; Database: D1 Menu Items; Components: Order Placement
FR-02.3	Allow waiter to add one or more menu items to order	DFD Level 2.1: Process 2.2 "Add/Update/Remove Menu Items"; Sequence Diagram: Place Order – loop for item selection; Classes: Order, OrderItem, MenuItem; Components: Order Placement
FR-02.4	Allow waiter to update or remove selected menu items before submission	DFD Level 2.1: Process 2.2 "Add/Update/Remove Menu Items"; Sequence Diagram: Modify or Cancel Order with modifications; Activity Diagram: Order Modification and Cancellation Workflow; Classes: Order, OrderItem, OrderManagementService; Components: Order Placement

FR ID	Functional Requirement	Design Mapping
FR-02.5	Allow customers to modify or cancel orders through waiter before submission	DFD Level 2.2: Process 2.9 "Process Order Cancellation"; Sequence Diagram: Modify or Cancel Order; Activity Diagram: Order Modification and Cancellation Workflow; Classes: OrderManagementService, Order; Components: Order Processing
FR-02.6	Automatically calculate 0.5% discount per 1,000 PKR when total $\geq$ 10,000 PKR	DFD Level 2.1: Process 2.3 "Calculate Discount"; Sequence Diagram: Place Order – calculateDiscount() with note "Discount = 0.5% per 1,000 PKR"; Activity Diagram: Order Workflow – discount calculation; Classes: OrderManagementService; Components: Order Placement
FR-02.7	Automatically add complementary items when total $\geq$ 10,000 PKR (up to 2% value)	DFD Level 2.1: Process 2.4 "Select & Add Complementary Items"; Sequence Diagram: Place Order – selectComplementaryItems() with note "System automatically selects complementary items up to 2% of subtotal"; Activity Diagram: Order Workflow – complementary item selection; Classes: OrderManagementService, ComplementaryItem; Database: D3 Complementary Items; Components: Order Placement

FR ID	Functional Requirement	Design Mapping
FR-02.8	Select complementary items from predefined list (drinks, desserts, appetizers)	DFD Level 2.1: Access to D3: Complementary Items; Sequence Diagram: Place Order - getActiveComplementaryItems(); Classes: ComplementaryItem, MenuManagementService; Database: D3: Complementary Items; Components: Order Placement
FR-02.9	Mark complementary items as free of cost	Sequence Diagram: Place Order – markAsComplementary(); Classes: OrderItem (isComplementary attribute), ComplementaryItem; Components: Order Placement
FR-02.10	Generate and display order summary with all details	DFD Level 2.1: Process 2.6 "Generate Order Summary"; Sequence Diagram: Place Order – getOrderSummary(); Classes: Order, Bill; Components: Order Placement
FR-02.11	Forward confirmed order to cashier and kitchen staff	DFD Level 2.1: Process 2.7 "Confirm & Submit Order"; DFD Level 2.2: Order flow to cashier and kitchen; Sequence Diagram: Send Order to Kitchen – sendOrderToKitchen(), forwardOrderToCashier(); Activity Diagram: Order Workflow – Step 8 "Forward Order to Cashier & Kitchen"; Classes: OrderManagementService, KitchenService, BillingService; Components: Order Processing

FR ID	Functional Requirement	Design Mapping
FR-03.1	Assign unique order ID to each confirmed order	Sequence Diagram: Send Order to Kitchen – assignUniqueOrderId(); Activity Diagram: Order Workflow – Step 7 "Generate Unique Order ID"; Classes: Order (orderId attribute); Components: Order Processing
FR-03.2	Transmit order details to cashier and kitchen	DFD Level 2.2: Process 2.7 showing order transmission; Sequence Diagram: Send Order to Kitchen – complete transmission flow; Activity Diagram: Order Workflow – parallel activities; Classes: OrderManagementService, KitchenService, BillingService; Components: Order Processing
FR-03.3	Set initial status to "Order Pending"	DFD Level 2.2: Process 2.9 initial status; Sequence Diagram: Send Order to Kitchen – updateStatus(ORDER_PENDING); Activity Diagram: Order Workflow – initial status setting; Classes: Order, OrderStatus enum (ORDER_PENDING); Database: D4 Order Status; Components: Order Processing
FR-03.4	Allow modifications or cancellations before kitchen acceptance	DFD Level 2.2: Process 2.9 "Process Order Cancellation"; Sequence Diagram: Modify or Cancel Order – status validation; Activity Diagram: Order Modification and Cancellation Workflow – pre-acceptance flow; Classes: OrderManagementService, Order; Components: Order Processing

FR ID	Functional Requirement	Design Mapping
FR-03.5	Prevent modification or cancellation after kitchen acceptance	DFD Level 2.2: Process 2.8 "Process Order Modification" with lock status; Sequence Diagram: Modify or Cancel Order – status check preventing changes; Activity Diagram: Order Modification and Cancellation Workflow – "Not Allowed" alternative flow; Classes: OrderManagementService, Order; Components: Order Processing
FR-03.6	Allow additional items to be added after kitchen acceptance	Sequence Diagram: Modify or Cancel Order – loop for adding new items; Activity Diagram: Order Modification and Cancellation Workflow – "Add New Items" flow; Classes: OrderManagementService, Order; Components: Order Processing
FR-03.7	Track and update order status through lifecycle	DFD Level 2.2: Process 2.10 "Update Order Status"; Activity Diagram: Order Workflow – complete status transitions; Classes: Order, OrderStatus enum (ORDER_PENDING, IN_PREPARATION, PREPARED, SERVED, COMPLETED); Database: D4 Order Status; Components: Order Processing
FR-03.8	Display updated status to waiter, cashier, and kitchen	DFD Level 2.2: Status update flows; Sequence Diagram: Send Order to Kitchen – notifications; Classes: NotificationService; Components: Order Processing

FR ID	Functional Requirement	Design Mapping
FR-04.1	Display all pending orders to kitchen staff	Sequence Diagram: Send Order to Kitchen – display pending orders; Activity Diagram: Order Workflow – Step 9a "Kitchen Receives Order"; Classes: KitchenService, Kitchen, KitchenOrder; Components: Kitchen Operations
FR-04.2	Allow kitchen staff to accept assigned order	Sequence Diagram: Send Order to Kitchen – acceptOrder(), addToPendingOrders(); Activity Diagram: Order Workflow – Step 10a "Kitchen Staff Accepts Order"; Classes: KitchenService, KitchenOrder; Components: Kitchen Operations
FR-04.3	Update status to "In Preparation" when accepted	Sequence Diagram: Send Order to Kitchen – updateStatus(IN_PREPARATION); Activity Diagram: Order Workflow – Step 11a "Status = In Preparation"; Classes: Order, OrderStatus (IN_PREPARATION), KitchenService; Components: Kitchen Operations
FR-04.4	Prevent modifications or cancellations once accepted	DFD Level 2.2: Process 2.8 lock status mechanism; Sequence Diagram: Modify or Cancel Order – status validation; Activity Diagram: Order Modification and Cancellation Workflow – "Not Allowed" flow; Classes: OrderManagementService; Components: Kitchen Operations

FR ID	Functional Requirement	Design Mapping
FR-04.5	Allow kitchen to mark order as “Prepared”	Activity Diagram: Order Workflow – Step 13a “Mark Order as Prepared”; Classes: KitchenService, Kitchen; Components: Kitchen Operations
FR-04.6	Notify waiter when order marked as “Prepared”	Sequence Diagram: Send Order to Kitchen – notifyKitchenStaff(); Activity Diagram: Order Workflow – Step 14a “Notify Waiter Order Ready”; Classes: NotificationService; Components: Kitchen Operations
FR-05.1	Notify waiter when order is prepared	Sequence Diagram: Feedback submission showing notification flow; Activity Diagram: Order Workflow - Step 15 ”Waiter Receives Notification”; Classes: NotificationService, Waiter; Components: Order Serving
FR-05.2	Allow waiter to mark order as “Served”	Activity Diagram: Order Workflow – Step 16 “Waiter Marks Order as Served”; Classes: OrderManagementService, Waiter; Components: Order Serving
FR-05.3	Update status to “Served” when confirmed	Activity Diagram: Order Workflow – Step 17 “Update Status = Served”; Classes: Order, OrderStatus (SERVED); Components: Order Serving
FR-05.4	Allow waiter to mark order as “Completed”	Activity Diagram: Order Workflow – Step 19 “Waiter Marks Order as Completed”; Classes: OrderManagementService, Waiter; Components: Order Serving

FR ID	Functional Requirement	Design Mapping
FR-05.5	Update status to "Completed" and display to dashboards	Activity Diagram: Order Workflow – Step 20 "Update Status = Completed" with note "Order ready for billing"; Classes: Order, OrderStatus (COMPLETED); Components: Order Serving
FR-06.1	Generate bill with all details including discounts and complementary items	Activity Diagram: Billing & Payment Workflow – Steps 1–10; Activity Diagram: Order Workflow – Step 21 "Cashier Generates Bill"; Classes: BillingService, Bill (subtotal, discount, tax, serviceCharge, totalAmount); Components: Billing and Payment
FR-06.2	Allow cashier to select payment method (cash, card, online)	Activity Diagram: Billing & Payment Workflow – Step 12 "Display Payment Method Options"; Activity Diagram: Order Workflow – payment method decision point; Sequence Diagram: Payment processing method selection; Classes: PaymentGateway, PaymentMethod enum (CASH, CARD_SWIPE, ONLINE_BANK, DIGITAL_WALLET); Components: Billing and Payment
FR-06.3	Validate payments based on selected method	Activity Diagram: Billing & Payment Workflow – Steps 13a–23b showing validation flows; Sequence Diagram: Payment validation for each method; Classes: PaymentGateway, BillingService; Components: Billing and Payment

FR ID	Functional Requirement	Design Mapping
FR-06.4	Process payments securely and log transactions	Activity Diagram: Billing & Payment Workflow – Steps 14–19 payment processing and logging; Activity Diagram: Order Workflow – Step 26 "Log Transaction Details"; Sequence Diagram: Transaction logging flow; Classes: PaymentGateway, Payment; Components: Billing and Payment
FR-06.5	Generate payment receipt with complete details	Activity Diagram: Billing & Payment Workflow – Step 22 "Generate Payment Receipt"; Activity Diagram: Order Workflow – Step 25 "Generate Payment Receipt"; Classes: PaymentReceipt (receiptId, paymentId, orderDetails, amounts), BillingService; Components: Billing and Payment
FR-06.6	Store receipts and transactions for accounting/auditing	Activity Diagram: Billing & Payment Workflow – Step 26 "Log Transaction Details"; Activity Diagram: Order Workflow – Step 32 "Archive Payment & Maintain Records"; Classes: Payment, PaymentReceipt; Components: Billing and Payment

FR ID	Functional Requirement	Design Mapping
FR-06.7	Allow customer feedback submission after payment	Activity Diagram: Billing & Payment Workflow – Step 27 "Display Feedback Form (Rating 1–5)"; Sequence Diagram: Feedback submission sequence; Activity Diagram: Order Workflow – Step 27 "Display Feedback Form"; Classes: FeedbackService; Components: Customer Feedback
FR-07.1	Allow feedback submission after successful payment	Activity Diagram: Billing & Payment Workflow – Steps 27–29 feedback flow; Sequence Diagram: Complete feedback submission flow; Activity Diagram: Order Workflow – feedback steps; Classes: FeedbackService, Feedback; Components: Customer Feedback
FR-07.2	Provide rating scale 1–5	Activity Diagram: Billing & Payment Workflow – Step 27 "Rating 1–5"; Sequence Diagram: Rating scale interface; Classes: Feedback (rating: Integer); Components: Customer Feedback
FR-07.3	Store feedback with order information	Activity Diagram: Billing & Payment Workflow – Step 29c "Save Feedback with Order ID & Timestamp"; Sequence Diagram: storeFeedback() method; Classes: Feedback (orderId, rating, comments, timestamp); Components: Customer Feedback
FR-08.1	Generate detailed sales reports (daily, weekly, monthly)	Classes: ReportingService, SalesReport; Components: Administration

FR ID	Functional Requirement	Design Mapping
FR-08.2	Include order count, sales, discounts, complementary values, payment summaries	Classes: SalesReport (totalOrders, totalSales, totalDiscounts, complementaryValue, paymentBreakdown); Components: Administration
FR-08.3	Identify most-selling and low-performing items	Classes: ReportingService, MenuManagementService; Components: Administration
FR-08.4	Allow admin to add/update/remove menu items	Classes: MenuManagementService, Admin; Components: Administration
FR-08.5	Allow admin to manage complementary items list	Classes: MenuManagementService, ComplementaryItem; Components: Administration
FR-08.6	Allow admin to manage user accounts	Classes: UserManagementService (addUser, updateUser, deactivateUser, removeUser), Admin; Components: Administration
FR-08.7	Display error messages for invalid input/operations	Activity Diagram: Error handling flows in all workflows; Classes: All service classes with error handling; Components: All Modules

5 Github and Figma Link:

- **GitHub Repository:** https://github.com/muhammadshoaibnoor/POS-System
- **Figma Design Files:** https://www.figma.com/file/shuaibnoor/POS-System-Design